

Training in the Canadian Arctic

Those who are charged with enforcing building performance standards in Canada's newest territory have miles to go before they begin.

by Stewart Selman

At about 6 pm on November 26, 2002, I log onto the Internet and check the weather conditions in Iqaluit, Nunavut, Canada. Current temperature: -18°F. The sun rose at 8:27 am, and set at 2:13 pm. Oh, the low tonight will be -28°F, but no wind.

Nunavut is Canada's newest territory. It was created out of the Northwest Territories as a result of years of negotiations between the Inuit people and the central Canadian government. Located above the tree line, it is snow-covered land most of the year. The terrain is predominantly rocky tundra with stunted vegetation. Winters last nine months, with an average January temperature of -22°F. The average July temperature is 59°F. It is a generally windy area; the prevailing winds come from the northwest. Nunavut has over 17,500 heating degree-days (9,800 degree days below 18°C). The population of Nunavut is 28,000, or about one person per 30 square miles. Nunavut is desolate, interesting, and beautiful—and it is experiencing a building boom.

Construction happens all year. Builders set up lights to work, since the days can be very short. And the summer is very short. You have to wear lots of layers of clothes, and working conditions can be extremely difficult. Any diagnostic testing on buildings will have to take place all year long. Yes, that means blower door tests at -22°F!

In early June of 2002, I received a phone call from Tim Mac Leod, the senior technical officer for the Architectural and Building section of the Technical Services Division of the Department of Public Works for the Territorial Government of Nunavut (GN). Canadians love long titles. Tim's section is responsible for all buildings built for and owned by the GN, including staff housing. He was shopping around for test equipment and training for building inspectors.



The GN is in the process of developing performance standards. Tim was looking for someone to teach the skills needed to enforce the performance standards, and to teach the diagnostic skills needed to analyze problems with new and existing buildings. Very few inspections are conducted in Nunavut, because inspections have not been institutionalized, and because it is very expensive to travel to outlying building sites to perform the inspections.

Tim asked what training I could provide for him in Minnesota's Twin Cities. Three thoughts entered my mind: (1) It would make more sense to train in Nunavut on their building stock. (2) This was an opportunity for me to travel to the Arctic. (3) It would be gratifying to have a positive influence on the construction of Nunavut building stock. Fortunately, a contract for training was issued, and in mid-August 2002 I was off to the Canadian Arctic.

Construction in Nunavut

Much of the ongoing construction is taking place in Iqaluit, the capital of Nunavut. The population of Iqaluit is about 6,000 people. The GN construction program includes schools, hospitals,

government office buildings, and staff housing projects. The GN is responsible for paying the heating bills of its properties. In Iqaluit alone, that amounts to CAN\$1 million a year.

The primary source of heating and electricity production is fuel oil. The oil is delivered to most of the region by ship during the summer months, when the bays and harbors are free of ice. Fuel shortage emergencies require extremely expensive air delivery. The government's goal is to save up to 20% on heating fuel costs by using proper insulating and sealing techniques.

Currently there are no mandatory building inspections. There are no building inspectors. The government will spec out a building and rely on the contractors to do a good job. There is some limited visual examination of insulation, weather barriers, and vapor retarders. Unfortunately, at the time of inspection these details are often half covered by drywall or are still being installed. The financial cost, and the time spent getting out to the outlying communities (most of the territory), are both huge (see "Cold, Costly Travel"). Therefore, performance standards are needed to ensure that the building envelope is constructed according to spec.

Much of the construction work is contracted to firms located far to the south in the population centers of Canada. Callbacks are not a part of the building culture. Small details missed cause big problems later. For example, wind-driven fine particle snow can be driven through gaps in the outside weather barrier. The snow remains frozen until the weather warms up. Then, the snow melts, and the resulting moisture damage causes the Sheetrock ceilings to collapse.

One of the most striking aspects of building in the Arctic is the fact that every structure is built on pilings. The pilings are set into the permafrost with water (yes, water—not concrete). The water freezes in the permafrost and the piling is set. All structures, from a single-family house to a 10,000 ft² school building (see photo on p. 12), are constructed in this manner. And, the bottoms of all these structures are exposed to the Arctic weather (see “Current Extreme-Cold Construction Practices”).

Buildings are heated primarily with oil-fired, atmospherically vented hydronic systems. Some newer construction uses forced-air distribution systems. Based on anecdotal information, spillage and backdrafting of these combustion appliances would seem to be common. Heat distribution in buildings is also a problem. Apparently, it is not unusual for occupants to regulate the heat by opening the upper-story windows. Problems caused by very low indoor relative

humidity, and by moisture getting into walls and freezing, are also prevalent.

Training

My primary task was to provide training on the use of the blower door and an infrared (IR) camera, and how to interpret the results. Pressure diagnostics—in particular worst-case



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This is drilling a hole the Nunavut way in preparation for worse case draft testing.

draft testing—was also a priority.

I take a building science and building system approach in my training. My philosophy as a trainer is to spend time in the classroom only as needed. The in-field experience not only transforms the abstract into something more tangible, but also gives students the opportunity to deal with the unexpected. Basically, I teach my students to take the knowledge and skills acquired in the

classroom and apply them to events that stray from the protocol. I teach them to use the protocol as a guide to gather basic data, especially as it concerns performance standards. However, I teach them that stretching the protocol is often how solutions are achieved.

The building-as-a-system philosophy is critical in building performance. It is possible to improve one aspect of a building's performance while degrading another. In Nunavut, one of the main goals was to improve building envelope performance. Benefits would include lower energy costs, reduced damage from wind-driven snow, and increased occupant comfort. However, tightening the building envelope could also increase the risk of mechanically induced spillage of combustion products. By taking a building system approach, I teach my students to look at the different aspects of home performance and how these are interrelated, in order for them to see how best to approach a problem.

On Monday the official training began. In addition to Tim, three other members of the Buildings Division office participated in the training. Tim was the most familiar with the equipment, and he was a strong advocate of performance testing. There was some skepticism among the other students about the training, but once we ventured into the field and actually started using the equipment, the skepticism began to fade.

We had the benefit of being able to test recently completed but unoccupied buildings. Building occupants are

Cold, Costly Travel

Members of the Technical Services Division of GN's Department of Public Works commonly charter a plane so we can hit several communities in a day or two. We charter small planes because most communities have short gravel runways and large aircraft simply can't land or take off. Some communities may have only one or two scheduled flights per week. Going on a scheduled flight could mean being stuck for up to a week in order to perform a simple vapor barrier inspection.

A charter is cheaper when the costs of airline tickets, hotels, and wages plus overtime are considered.

A worst-case scenario would be a poly inspection in Cambridge Bay. You pay airfare to travel to Yellowknife, and the cost of an overnight and meals at a hotel. The next day you catch a connecting flight to Cambridge Bay. You perform an inspection and stay overnight in a hotel (CAN\$150–\$200 per night, even if you're sharing a room with someone you've never met). You fly back to Yellowknife the next day, stay overnight, and fly to Iqaluit

the following day. So the GN could potentially pay airfare (CAN\$2,400 round trip), salary for four days, three night's worth of accommodations, and all meals and expenses for what could be a four-hour inspection. Imagine if we performed the standard foundation, framing, insulation, vapor barrier, electrical, mechanical, and final inspections on every project in every community. The cost of proper inspections could easily surpass the cost to actually build the building.

—Tim MacLeod

always a problem. The first building we tested was a town house block consisting of 16 units situated side by side and constructed by a private contractor on speculation with the intention of selling the property to the GN.

Gaining the Experience

The focus of the first day's training was going to be blower door setup and operation, and running automated tests with the APT and TECTITE software. Although temperature conditions in August were not ideal, infrared scanning was incorporated into the protocol. The protocol was very basic. We set up the blower door, ensured that the heating system and water heater did not turn on during the test, and conducted a base IR scan. This base scan notes insulation voids and provides a comparison to the post-blower-door scan. Then we ran an automated blower door test while checking for leaks with a smoke bottle, and conducted a post-blower-door scan to identify bypasses.

The blower door used with a smoke bottle can be a very effective tool. It is important to take a visual survey of the building and try to understand how it is put together. The more experience you have with a particular building style, the easier this becomes. Nevertheless, an auditor needs to be prepared for surprises. These particular town house-style residences were constructed with a complicated multilevel design. The mechanical room included an oil-fired water heater and furnace in addition to a washer and dryer. Adjacent to this room was a low-headroom crawlspace that shared part of a common wall with the mechanical room. Ductwork passed through a chaseway connected to this same wall. There were also a number of penetrations through the floor of the crawlspace, including combustion air supply and some plumbing.

We set up the equipment in an end unit. The space was approximately 1,100 ft² (volume approximately 10,000 ft³). The multipoint blower door test result was 1,173 CFM₅₀. Some very crude zonal pressure diagnostics indicated some connection to the adjacent unit, but the connection was not quantified. The 1 CFM₅₀/ft² leakage

was much too high. (There is no standard for leakage at this point, but in the future there will be a call to meet the Canadian R-2000 standards, which allow no more than about 0.25 CFM₅₀/ft² leakage.) When we isolated the crawlspace as well as we could from the rest of the unit, we found that the



This Minneapolis Blower Door canvas allows the insertion of two fans—for larger buildings with large leaks.

crawlspace was half in and half out of the structure. Much of the leakage to the outside was visually identified through unsealed bypasses around combustion air openings and floor-wall seams. The best remedy that could be hoped for in this situation would be sealing accessible leaks with foam insulation or caulk. (Since the GN did not own the building at the time we tested, it is unclear if these fixes will be made.)

These defects in the building envelope increase energy costs and can lead to inconveniences, such as boxes freezing to the floor of the crawlspace. More serious problems, such as Sheetrock blowing apart from moisture being driven into it and then expanding, are also a concern. Another Arctic bypass effect occurs when moisture is pushed through small penetrations in the exterior envelope in high moisture areas, such as bathrooms. This can create ice balls in the walls. These ice balls will keep expanding until they blow the siding right off of a building.

Significant leakage was also identified at the chaseway that went to the space below the roof area. Addressing the leaks at this point is very difficult and in many cases not practical. Performance standards would make it easier to call a contractor to task on missed details that produce leaks of this kind.

The temperature conditions in mid-August were not good for IR camera scanning. There was not enough contrast between indoor and outdoor temperatures, which meant that insulation voids and bypasses were not clearly defined on the IR scans. (However, conditions for IR scanning are very good most of the year.) We did go through the IR protocol for training purposes. I am certain that the infrared imagery and other test results will be enlightening to contractors who will swear that every last specified detail was met.

Another residential structure that we trained at was a three-story fourplex that was divided by a sealed wall down the middle. Testing indicated that the separation was quite good. Each side was divided into two residences. The upper two stories made up one residence, and the first story consisted of a very small apartment, a laundry room, and the mechanical room. The mechanical rooms in these units were large. They included a tank for potable water, a wastewater tank, and an oil-fired water heater and furnace.

We conducted testing similar to that described above. We also established a worst case depressurization draft protocol (see "Thorough Diagnostic Testing of Vented Appliances," *HE* Sept/Oct '99, p. 43 for a description of a similar protocol). The worst case draft protocol consists basically of establishing base pressure readings in the combustion equipment venting with respect to the combustion appliance zone (CAZ), and in the CAZ with respect to outside. Mechanical devices (exhaust fans, dryer, air handling systems) are operated and doors are positioned to create the greatest depressurization in the CAZ. The principal combustion appliance being tested—for example, a water heater—is fired up, measurements are recorded and spillage is checked with smoke, mechanical devices are turned off, and pressures are recorded again. When we

Current Extreme-Cold Construction Practices

Buildings are built on pilings and raised above ground level by 2 ft or more, allowing air to blow freely under the building. This prevents the building from insulating the permafrost from the outdoor air and prevents building heat from melting the permafrost and losing its stability. When this happens, the buildings begin to sink!

A typical floor system consists of

1. pilings,
2. wooden or steel beams,
3. a plywood soffit with battens covering joints,
4. 1-inch rigid insulation,
5. a Tyvek air barrier,
6. floor joists insulated with R-40 batt insulation, and
7. a plywood subfloor.

Typical exterior wall construction consists of

1. wood or metal siding,
2. 1 x 3 wood strapping,
3. 1-inch rigid insulation,
4. a Tyvek air barrier,
5. standard exterior sheathing,
6. 2 x 6 wood studs,
7. R-20 fiberglass batt insulation,
8. a continuous 6-mil poly vapor retarder,
9. 2 x 2 horizontal wood strapping, and
10. Sheetrock.

The 2 x 2 horizontal wood strapping allows electrical boxes, wiring, and piping to be installed on the inside of the poly without penetrations. The gap created by the 2 x 2s can be left as an air space or filled with

1-inch rigid or semirigid insulation.

Typical roof construction (usually sloped) consists of

1. metal roofing,
2. a Tyvek air barrier,
3. standard exterior sheathing,
4. roof joists,
5. R-40 fiberglass batt insulation,
6. a 6-mil poly vapor retarder,
7. 2 x 2 wood strapping, and
8. Sheetrock.

The entire building is wrapped and sealed with Tyvek. There is no venting, due to the extremely fine snow. During a blizzard the high winds can drive this fine snow and easily fill a vented attic space. It is common for ceiling drywall to collapse in the spring from the weight of melting snow that has accumulated in attic spaces over the winter. This snow is blown in, not through attic venting, but through gaps in an improperly sealed air barrier. Even a small penetration can be disastrous. Once I arrived at a new home construction site after an overnight blizzard to find 4 ft of snow in the kitchen that had blown in through a 2-inch diameter precut opening for a deadbolt in an exterior door.

Another problem that occurs in extreme-cold conditions is damage to the poly vapor retarder from flexing caused by wind pressure. More and more, we are using the exterior sheathing as the vapor retarder. All the joints in the sheathing are sealed with 4-inch to 6-inch-wide strips of modified bituminous membrane, such as is used in flat roofing.

Rigid insulation is hung on the exterior of the building. The stud cavities are not insulated, and wiring and plumbing may

be run without affecting the envelope. In many cases less-expensive 2 x 4 exterior framing can be used instead of 2 x 6 framing, because the R-20 fiberglass batts are no longer needed. This also saves on material and shipping costs. Every little bit helps, because all materials must be transported to the communities by ship or air after they are purchased down south.

I am looking into building with structural insulated panels (SIPs). The panels are custom-fabricated at the factory and assembled on-site. With proper installation, the end result can be a better insulated and air/vapor sealed building than can be achieved using standard construction methods. (The climate around Nunavut is relatively dry. For a cautionary note on SIPs building in a wetter climate, see "A Serious Challenge to a New Building Technology," *HE* Jan/Feb '02, p. 7.) The insulation, vapor barrier, and sheathing are complete as soon as the panels are assembled, allowing efficient temporary heating early in the construction stage. Blower door tests on this type of construction show exceptionally tight buildings. Construction labor costs can be substantially reduced using this system. Energy savings can also be realized, because there is less air infiltration of the building envelope. Information can be found at www.plastifab.com. This could be the answer to many problems caused by extreme temperatures (construction continues year round, even at -30°F), and by the special problems that we face in sealing buildings.

—Tim MacLeod

conducted this test, the CAZ depressurized by 10 Pa. The vent was positive before the water heater was turned on, but established draft quickly enough (within 30 seconds) so that this initial positive pressure was not a problem. Temperatures are 70°F colder in Iqaluit today than they were on the day of the training. It is quite possible that a colder vent could increase the time necessary to establish a draft. If the vents were really cold and there was a steady down-draft through the vent, establishing a

draft would be even more difficult.

Typically one does not open windows as part of a worst case draft test. Earlier I mentioned that in winter (most of the year), the building occupants open the upper-story windows to regulate the heat. We tested to see how much depressurization we could get in the CAZ if we opened a window on the leeward side of the wind. We got 6–7 Pa just from the open window. There is a good chance that some of the combustion product

spillage that is being experienced could be due to the opening of windows. (Unfortunately, this is a problem the Arctic shares with the lower 48.) If other improvements in the building envelope result in better heating distribution and improved comfort so that the windows are not opened, perhaps some of the spillage issues will go away.

The worst case draft testing was documented and facilitated by the use of APT, with a laptop computer and software that was written by myself. The

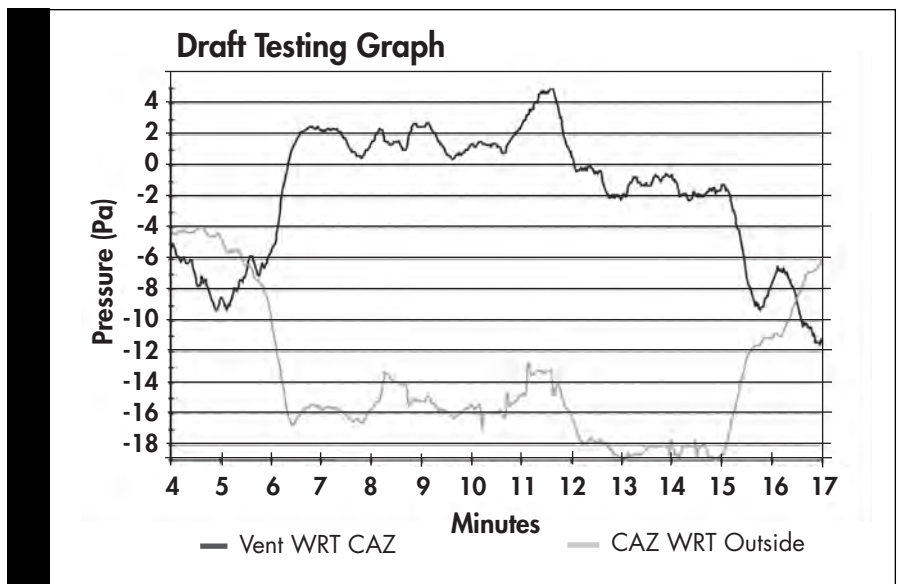


Figure 1. The big dip in the CAZ WRT outside pressure is primarily from turning the kitchen exhaust fan on (at around five minutes). The dryer and bathroom fans had minimal impact. As the house depressurizes the vent goes towards positive and in this case above zero. At about 10.5 minutes the water heater is turned on and is spilling. It starts to draft at about 12 minutes but it is a weak draft. The kitchen exhaust fan is turned off at about 15 minutes. Notice how the house becomes less negative to the outside while the vent becomes more negative to the CAZ. This is a good example of mechanically induced depressurization and its effect on combustion appliance draft.

entire process was graphed (see Figure 1). Graphing is a great documentation tool, and it is useful in solving pressure problems. It is also a wonderful training instrument, even if the trainees are not going to be using an APT. Graphing of pressure relationships takes an invisible and abstract measure, such as 10 Pa of depressurization, and places it into a visual format. When trainees see the CAZ/outside line going down (more negative) while the vent/CAZ line is going up (more positive), it helps them to internalize the concept.

The last building that we trained in was a new school. The finished floor area was 7,000 ft². There was an unfinished lower-level crawlspace under the building. This building was going to present us with the challenge of using two blower door fans at once. We expected that there would be enough leaks that one fan would not be enough. For this reason, Tim had purchased two fans and a Minneapolis Blower Door canvas that allows the insertion of two fans (see photo on p. 14). The largest buildings that GN will be dealing with are rarely over 10,000 ft².

We experimented with a variety of

ways to use the two fans at once. We found that the best way is to crank one fan all the way up to maximum and then use the second fan to give you the extra exhaust flow to get to -50 Pa (after considering base pressures). The school measured out at around 5,200 CFM₅₀ (much larger than it should have been, but not as bad as it could have been). Inspection with the blower door running and smoke identified many large and small bypasses that should have been addressed at the time of construction, and weren't.

Details, Details

Testing for building performance is an issue that has been written about in *Home Energy* for 20 years. If you don't test, you don't know. For those contractors who do want to provide their best product, performance testing is a wonderful educational tool. For those contractors who don't care, performance standards with testing carry the cost of making it right. Many of the challenges builders and inspectors face in Nunavut are the same ones that their counterparts face farther south. The same lack of

attention to details that causes problems in the colder regions of the United States is causing problems in the Canadian Arctic. But because of the harsh conditions, the building damage that is caused by missed details may appear more quickly in the Arctic.

From a trainer's perspective, I had a wonderful experience. I had the privilege of working with motivated students and I had good buildings to teach with. Small class sizes allowed for a lot of hands-on training and exchange of ideas. We were able to discuss methods of solving building performance problems. My basic ideas on how to use the equipment and how to think of building performance were stretched and challenged, which is a very good thing. Hey, I also got to see the beautiful Canadian Arctic, stand in the depths of Hudson Bay during low tide, eat fresh caribou, meet some great people, and have an all-around memorable experience.

Stewart Selman of Stewart Selman & Associates, when not training in the Arctic, does house diagnostics in the Twin Cities area of Minnesota. Tim MacLeod, the senior technical officer for the Architectural and Building section of the Technical Services Division of the Department of Public Works for the Territorial Government of Nunavut, contributed greatly to this article.

For more information:

Information about training programs for home performance professionals, including Stewart Selman & Associates' cold-weather offerings, will appear in the July/August '03 issue of *Home Energy*.

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